

Angular Reactive Forms

<https://angular.dev/guide/forms/reactive-forms>

Overview

- ◆ Reactive forms provide a approach to handling **form inputs whose values change** over time.
 - create and update a basic form control
 - use multiple controls in a group
 - validate form values
 - create dynamic forms
- ◆ Reactive forms use an approach to **managing the state** of a form

Adding a basic form control

1. Import the ReactiveFormsModule
2. Generate a new component with a FormControl
3. Register the control in the template
4. Display the component

Adding a basic form control

```
import { ReactiveFormsModule } from '@angular/forms';

@Component({
  selector: 'app-name-editor',
  standalone: true,
  imports: [ReactiveFormsModule],
  templateUrl: './name-editor.component.html',
  styleUrls: ['./name-editor.component.css']
})
```

```
export class NameEditorComponent {
  name = new FormControl('');

  updateName() {
    this.name.setValue('Nancy');
  }
}
```

```
<label for="name">Name: </label>
<input id="name" type="text" [formControl]="name">

<p>Value: {{ name.value }}</p>

<button type="button" (click)="updateName()">Update Name</button>
```

Grouping form controls

1. Create a FormGroup instance
2. Associate the FormGroup model and view
3. Save form data
4. Display the component

Grouping form controls

```
import {ReactiveFormsModule, FormGroup, FormControl} from '@angular/forms';

@Component({
  selector: 'app-profile-editor',
  standalone: true,
  imports: [ReactiveFormsModule],
  templateUrl: './profile-editor.component.html',
  styleUrls: ['./profile-editor.component.css']
})
export class ProfileEditorComponent {
  profileForm = new FormGroup({
    firstName: new FormControl(''),
    lastName: new FormControl(''),
  });

  onSubmit() {
    // TODO: Use EventEmitter with form value
    console.warn(this.profileForm.value);
    console.warn(this.profileForm.value.firstName);
    console.warn(this.profileForm.value.lastName);
  }
}
```

```
<form [formGroup]="profileForm" (ngSubmit)="onSubmit()">
  <label for="first-name">First Name: </label>
  <input id="first-name" type="text" formControlName="firstName">
  <p></p>
  <label for="last-name">Last Name: </label>
  <input id="last-name" type="text" formControlName="lastName">
  <p><button type="submit">Submit</button></p>
</form>
```

Creating nested form groups

- ◆ Create a nested group.
- ◆ Group the nested form in the template.
- ◆ Updating parts of the data model

First Name:

Last Name:

Address:

Street: District: City:

Validating form input

- ◆ Import a validator function
- ◆ Make a field required
- ◆ Display form status

```
import {ReactiveFormsModule, Validators, FormGroup, FormControl} from '@angular/forms';
```

```
export class ProfileEditorComponent {  
  profileForm = new FormGroup({  
    firstName: new FormControl('', Validators.required),  
    lastName: new FormControl(''),  
    address: new FormGroup({  
      street: new FormControl(''),  
      district: new FormControl(''),  
      city: new FormControl('', Validators.required),  
    }),  
  });  
});
```

<p>Complete the form to enable button.</p>

```
<button type="submit" [disabled]="!profileForm.valid">Submit</button>
```